

Aim/Overview of the practical: (B). To design the circuit and write an Arduino program to interface an SG90 servo motor with Arduino Uno and control its angular position automatically, while displaying the servo angle on a 16×2 LCD with I2C Communication.

COMPONENTS REQUIRED

S.No.	Component	Specification	Quantity
1	Arduino Uno board	ATmega328P	1
2	SG90 micro servo motor	4.8–6 V, 0°–180°	1
3	16×2 LCD with I2C module	PCF8574, address 0x27 / 0x3F	1
4	Breadboard	Standard	1
5	Jumper wires	Male–male / male–female	As required
6	External 5 V supply (optional)	For servo under load	1
7	USB cable (Type A–B)	For programming	1
8	Computer with Arduino IDE	Servo and LiquidCrystal_I2C libraries	1

THEORY

1. Servo Motor

A Servo motor is a rotary actuator that allows precise control of angular position, usually within a range of 0° to 180°. It consists of a small DC motor, a gear-reduction assembly, a position-feedback potentiometer and a built-in control circuit, all housed together in a single unit. The feedback potentiometer continuously reports the current shaft position to the internal control circuit, which drives the motor until the shaft reaches the commanded angle - this closed-loop action is what makes a servo hold a precise position instead of just spinning continuously like a normal DC motor.

The SG90 servo has three pins that are typically labelled as follows:

- **VCC (Power Supply):** This pin is connected to the power supply of the circuit, typically 5V.
- **GND (Ground):** This pin is connected to the ground of the circuit.
- **Signal (Control):** This pin is used to control the position of the servo motor. A pulse-width modulated signal is applied to this pin, with the width of the pulse determining the position of the motor. The pulse width must be between 600us and 2400us to rotate the motor from 0° to 180°.

Why No Motor Driver IC Is Needed

Unlike a plain DC motor, a Servo motor already contains its own internal driver circuit that supplies the drive current to its small internal motor. Because of this, the Servo's signal wire only needs to carry a low-current logic-level PWM signal, which an Arduino digital pin can supply directly - so, unlike a DC motor, a Servo does not need an external motor-driver IC such as the L293D. A small hobby servo (SG90/MG90) can even be powered

directly from the Arduino's 5V pin; a larger servo should instead be powered from a separate 5-6V supply with its ground connected common to the Arduino.

PIN CONNECTIONS

SG90 → Arduino Uno

SG90 Wire	Arduino Uno	Function
Brown/Black	GND	Ground
Red	5V	Power
Orange/Yellow	D9	Servo signal

I2C LCD → Arduino Uno

I2C LCD	Arduino Uno
VCC	5V
GND	GND
SDA	A4
SCL	A5

16×2 LCD with I2C Module

A 16×2 LCD displays 16 characters on each of 2 lines. The I2C backpack (PCF8574) reduces the wiring to two data lines, SDA and SCL, plus power. On Arduino Uno, SDA is pin A4 and SCL is pin A5. The LCD is controlled with the LiquidCrystal_I2C library; its address is usually 0x27 (or 0x3F).

Automatic Control

In this practical the servo moves automatically without any user input: the program sweeps the shaft from 0° to 180° and back to 0° in 1° steps, repeating continuously, while the present angle is shown on the LCD.

Automatic Working

The Arduino automatically moves the SG90 servo through different angular positions:

0° → 90° → 180° → 90° → 0° → Repeat

At each position, the **16×2 I2C LCD displays the current angle.**

WORKING

The Arduino generates the control signal for the **SG90 servo motor through D9**. The Servo library is used to command the servo to specific angular positions. The servo automatically moves from **0° to 90°, then to 180°, and back to 90°**.

At every position, the **16×2 I2C LCD displays the corresponding servo angle**. After completing the sequence, the program repeats it continuously.

PROGRAM - SERVO MOTOR CONTROL with LCD

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

Servo myServo;
// I2C LCD address
LiquidCrystal_I2C lcd(0x27, 16, 2);
int servoPin = 9;

void setup()
{
    myServo.attach(servoPin);
    lcd.init();
    lcd.backlight();
    lcd.setCursor(0, 0);
    lcd.print("SG90 SERVO");
    lcd.setCursor(0, 1);
    lcd.print("INITIALIZING");
    delay(2000);
    lcd.clear();
}

void loop()
{
    // 0 Degree
    myServo.write(0);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("SERVO MOTOR");
    lcd.setCursor(0, 1);
    lcd.print("ANGLE: 0 DEG");
    delay(1000);
```

// 90 Degree

```
myServo.write(90);  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("SERVO MOTOR");  
lcd.setCursor(0, 1);  
lcd.print("ANGLE: 90 DEG");  
delay(1000);
```

// 180 Degree

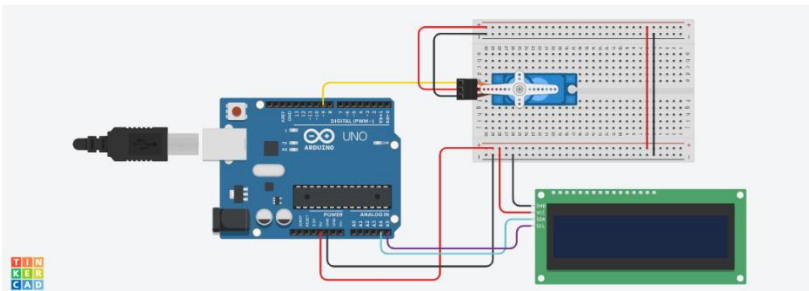
```
myServo.write(180);  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("SERVO MOTOR");  
lcd.setCursor(0, 1);  
lcd.print("ANGLE: 180 DEG");  
delay(1000);
```

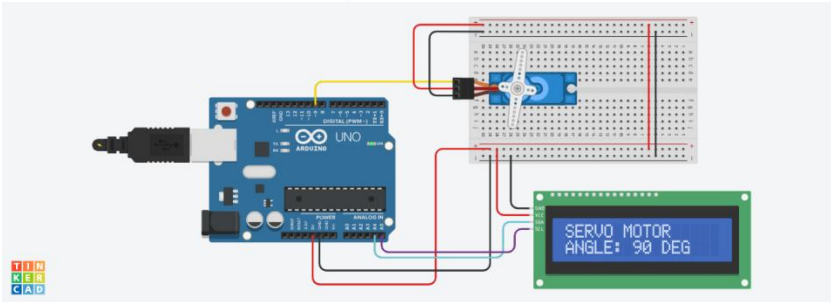
// Return to 90 Degree

```
myServo.write(90);  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("SERVO MOTOR");  
lcd.setCursor(0, 1);  
lcd.print("ANGLE: 90 DEG");  
delay(1000);
```

}

OUTPUT





PROCEDURE

- Place the Arduino Uno, SG90 servo motor, and 16×2 I2C LCD on the workbench.
- Connect the signal wire of the SG90 servo to Arduino digital pin D9.
- Connect the servo VCC to 5V and GND to GND of the Arduino.
- Connect the I2C LCD:

SDA → A4

SCL → A5

VCC → 5V

GND → GND

- Connect the Arduino Uno to the computer using the USB cable.
- Open the Arduino IDE and include the required Servo, Wire, and LiquidCrystal_I2C libraries.
- Enter and verify the given Arduino program.
- Select the correct Arduino Uno board and COM port, then upload the program.
- After uploading, observe the SG90 servo movement.
- The servo automatically moves to 0°, 90°, 180°, and 90° positions.
- Observe the 16×2 I2C LCD, which displays the corresponding servo angle.
- Verify that the sequence repeats continuously.

OBSERVATION TABLE

S. No.	Servo Angle	LCD Display	Servo Position
1	0°	ANGLE: 0 DEG	Initial position
2	90°	ANGLE: 90 DEG	Middle position
3	180°	ANGLE: 180 DEG	Maximum position
4	90°	ANGLE: 90 DEG	Returns to middle
5	0°	ANGLE: 0 DEG	Sequence repeats

RESULT

The SG90 servo motor was successfully interfaced with the Arduino Uno and 16×2 I2C LCD. The servo motor was automatically controlled at different angular positions of 0°, 90°, and 180°, and the corresponding angle was successfully displayed on the LCD.

PRECAUTIONS

- Check the SG90 servo wire connections before powering the circuit.
- Connect the servo signal wire to D9 as specified in the program.
- Ensure that Arduino GND and LCD GND are properly connected.
- Do not force the servo horn beyond its mechanical limit.
- Avoid mechanically loading the SG90 beyond its rated capacity.
- Use a suitable 5V supply for the servo; if the servo causes Arduino resets or unstable operation, use a separate suitable 5V supply with common ground.
- Check the I2C LCD address. The common address is 0x27, while some modules use 0x3F.
- Do not make or change connections while the circuit is powered.
- Ensure that the jumper wires are firmly connected.
- Keep the servo and Arduino circuit away from short circuits and loose connections.